

Paul Dubois

pauldubois98@gmail.com | Paris, France | [LinkedIn](#)

SUMMARY

“leveraging science for a better world”

Ph.D. in Applied Mathematics (Artificial Intelligence applied to healthcare).

Expert in Data Science / Machine Learning. Proficient in coding and infrastructure management.

ACADEMIA

CentraleSupélec, Paris-Saclay, France
2026

September

Tenure Track, Département de Mathématiques
Researcher in Cancer Data Science Unit

CentraleSupélec, Paris-Saclay, France

December 2024

PhD, Applied Mathematics

Thesis title: “Automation of treatment planning for radiotherapy using AI optimisation techniques”

Oxford University, Oxford, United Kingdom

July 2021

Master of Science, Mathematical Science

GPA: 68% (merit)

Dissertation on Random Fractals and Branching Processes.

Key courses: Analytic Topology, Category Theory, Approximation of Functions, Theories of Deep Learning, Networks, Random Matrix Theory,

University College London, London, United Kingdom

July 2020

Master of Science Integrated, Mathematical Science

GPA: 85% (first)

Research project on Modular forms mod 2: “Governing Fields for the Hecke Algebra”

Key courses: Probability, Measure Theory, Spectral Theory, Functional Analysis, Multivariable Analysis, Differential Geometry, Analytic Number Theory, Graph Theory and Combinatorics, Elliptic Curves, Commutative Algebra, High-Performance Computing, Evolutionary Games and Population Genetics

INDUSTRY EXPERIENCE

PRF Development, Paris, France

October 2021 – Present

AI Engineer & Data Scientist

- **Client: Polytechnique / Inria CEDAR Team** (<https://team.inria.fr/cedar/>)
Developed a pipeline tool for anomaly detection on temporal graphs and time-series data.
- **Client: City University of New York** (<https://www.cuny.edu/>)
Developed AI tool for information retrieval.
- **Client: Wheere** (<https://www.wheere.com/>)
Trained developed AI models for signal processing; significantly improved stability of the system.
- **Client: Surgitec** (<https://www.surgitec.com/>)
Trained models for the segmentation of sensitive regions in the brain using MRI (identified by surgeons as “applicable for clinical treatment”)
 - o Developing AI software for Deep Brain Stimulation (DBS)
 - o Designed a cost function for ranking candidate trajectories for DBS
 - o Created a new algorithm for best trajectory selection
- **Web scrapping for Acceor** (<https://www.acceor.com/>)
- **Web scrapping Economics researcher from TSE**

RoomFurnisher – London, UK

July 2020 – September 2020

Engineer (Intern)

- Developed the startup's proof of concept by integrating open-source 3D reconstruction tools. Used **Python** and **AliceVision (Meshroom)** to perform **photogrammetry**, generating accurate 3D models of indoor environments from image datasets.

Airbus – Toulouse, France

June 2019 – September 2019

Data Scientist (Intern)

- Developed an internal **performance monitoring dashboard** from scratch using **SQL**, **HTML/CSS**, and **JavaScript** to visualize team productivity and key performance indicators (KPIs) metrics.
- Designed and deployed **Python-based automation tools** to streamline data collection and reporting workflows, reducing manual effort and improving data reliability across departments.

TEACHING

CentraleSupélec University, Paris, France

Engineering school

Lecturer

January - March {2023, 2024, 2025}

- Deep Learning & Natural Language Processing:
Deep learning from scratch (back-propagation), perceptron, convolutions, optimizers & learning rates, RNN, U-net, V-net, GANs; with Kaggles created as homework.
Each class contains approximately 35 students (per semester).
Students' Feedback: "Ranked as No.1 among the 3rd-year courses" (2025)

Teaching Assistant

Coding	Fall {2021, 2022, 2024}
Algorithms	Fall {2023, 2024}
Reinforcement Learning	Fall 2024
Informatics	Fall 2023
NLP project	Fall 2022
Computer vision	Spring 2022
Optimization	Fall 2021

Essec University, Cergy, France
2026}

September {2021, 2023, 2024, 2025,

Business school

External lecturer

- Mathematics:
Basic methodologies for proofs; linear algebra, differential calculus, integration, and asymptotic analysis; prerequisites of the courses in the master's.
Average satisfaction > 4,75/5 every year

Bootcamp Ironhack Barcelona, Spain

July 2023 - August 2023

Private school

Lead teacher/lecturer

- Data Science:
Basic data science using Python, SQL, web technologies, Tableau and basic mathematics. *The content covers the premises of programming, data scraping, mathematical modelling, and artificial intelligence. The main resources are property of IronHack, but extra resources (mainly on AI, created by myself) are open source.*

PUBLICATIONS & POSTERS

Research Interest: AI, Front-end AI, Computer Vision, National Language Models

Google Scholar: [Paul Dubois - Google Scholar](#)

1. **P. Dubois**, P. Fenoglietto, P.H. Cournède, N. Paragyos (2024). Radiotherapy Dose Optimization via Clinical Knowledge-based Reinforcement Learning. Artificial Intelligence in Medicine (AIME), (Salt Lake City, Utah, USA). https://dl.acm.org/doi/10.1007/978-3-031-66538-7_16
2. **P. Dubois**, P. Fenoglietto, P.H. Cournède, N. Paragyos (2024). Clinically Dependent Fully Automatic Treatment Planning System. American Society for Radiation Oncology (ASTRO), (Washington, DC, USA). [https://www.redjournal.org/article/S0360-3016\(24\)02860-8/fulltext](https://www.redjournal.org/article/S0360-3016(24)02860-8/fulltext).
3. **Paul Dubois**, Nikos Paragios, Paul-Henry Cournède, Gizem Temiz, Rafael Marini-Silva, Norbert Bus, Pascal Fenoglietto (2024). A Novel Framework for Multi-Objective Optimization and Robust Plan Selection Using Graph Theory. <https://user-swndwmf.cld.bz/ESTRO-2024-Abstract-Book/3687/> European Society for Radiotherapy and Oncology (ESTRO), Glasgow, UK.
4. **P. Dubois**, P. Fenoglietto, P.H. Cournède, N. Paragyos. (2024). Dose Volume Histograms Guided Deep Dose Predictions. Société Française de Physique Médicale (SFPM), Dijon, France
5. **P. Dubois**, P. Fenoglietto, P.H. Cournède, N. Paragyos. (2024). Attention Mechanism on Dose-Volume Histograms for Deep-dose Predictions. Société Française de Radio-Oncologie (SFRO), La Défense, France.

AWARDS, HACKATHONS & COMPETITIONS

- **Science&Vie Junior Innovation Prize, February 2017** (1000€) - Morse translating machine.
- **Top 10 Prize, July 2017** - at the University College of London in mathematics.
- **Porticode 3.0 Fortnightly, November 2018** – Prize winner for best 90's-style website.
- **MakeHarvard 3D Modeling Helper, February 2019** – Prize winner; hardware translating real-world rotations to 3D software.
- **Ceteris Paribus Face Challenge, April 2022** – 3rd place; applied transfer learning to facial feature modification.
- **Hackathon ISEP Text-to-Speech, May 2022** – Winner; developed state-of-the-art text-to-speech AI.
- **Hackathon ISEP Green Code, October 2022** – Winner; optimized Python program for energy efficiency.
- **Hackathon ISEP FrHack, November 2022** – 3rd place; discovered statistically significant correlations between phone signals and radio waves.
- **Hackathon Essonne Eau & Numérique, April 2023** – 2nd place; simulated water levels in images using AI-generated depth maps.
- **Hackathon InterCentrales Leaf Nothing Behind, April 2023** – 4th place; predicted vegetation on occluded images with AI.
- **Citadel C1games Terminal Competition, May 2023** – 3rd place; coded AI strategy for competitive game tournament.
- **Hackathon ISEP IT4Green, January 2024** – Winner of 'Refurnished' prize; eco-friendly digital technology solutions.

- **Hackathon ISEP FrHack, *March 2024*** – 2nd prize; optimized frequency allocation for wireless cameras.
- **Hackathon Georgetown HoyaHack, *January 2025*** – Winner of Databricks prize; participated in Major League Hacking event.